CLINICAL · APPROACH TO

Approach to facial pain and headache

Patients and doctors alike blame the sinuses for facial pain, but most people referred with "sinus headache" do not have sinus disease. Rhinosinusitis rarely causes pain without nasal blockage, discharge or smell loss, and in the absence of pus or inflammation on endoscopy the diagnosis is very unlikely. The approach is to take a structured pain history, examine the nose with an endoscope, avoid CT as a screening test, and recognise the primary headaches and neuralgias that are the real causes for most patients: migraine, tension-type headache, midfacial segment pain, the trigeminal autonomic cephalalgias and trigeminal neuralgia.

AT A GLANCE

- No nasal symptoms and normal endoscopy: it is not sinusitis.
- Midfacial segment pain responds to low-dose amitriptyline.
- Trigeminal neuralgia: carbamazepine and MRI.
- Numbness with pain = perineural tumour until imaged.

1 CHAPTER ONE The pain history

Most of the diagnosis.

1.1 A structured history

Use the same headings for every patient. **Site**: unilateral or bilateral, and the dermatome. **Character**: throbbing, pressing, stabbing or burning. **Severity**, out of ten: migraine and cluster headache often reach ten. **Timing**: how long each attack lasts, how often attacks come, and whether there are clusters and remissions. **Associated features**: nausea, photophobia and aura (migraine); a red, watering eye, rhinorrhoea, blockage and a drooping lid on the side of the pain (trigeminal autonomic cephalalgias); nasal symptoms (rhinosinusitis). **Triggers**: alcohol in cluster headache, touch or chewing in trigeminal neuralgia, the menstrual cycle in migraine. **Relieving factors** and what has been tried, including analgesic overuse, which itself causes daily headache.

Rhinosinusitis pain accompanies an attack of nasal symptoms and settles when they do. Pain that has not changed with years of antibiotics and nasal sprays is unlikely to be sinogenic.

RED FLAGS IN HEADACHE AND FACIAL PAIN

- **Thunderclap headache**, new headache with fever, neck stiffness, confusion or focal neurological signs: emergency imaging and neurology.
- **New headache over 50 with jaw claudication, scalp tenderness or visual symptoms:** giant cell arteritis — ESR, CRP and steroids the same day.
- **Headache worse lying down, on waking or with straining, with visual obscurations or pulsatile tinnitus:** raised intracranial pressure.
- **Facial numbness, diplopia, proptosis, a loose tooth or a nasal mass:** sinonasal malignancy.
- **Acute sinusitis with periorbital swelling, reduced vision or severe headache:** orbital or intracranial complication.

CHAPTER TWO

2 Examination and tests

Endoscopy first.

2.1 The examination

Examine the nose with an endoscope: mucopus or oedema in the middle meatus supports rhinosinusitis, and its absence makes it very unlikely. Examine the cranial nerves, especially the trigeminal nerve and the eyes, the temporomandibular joints, the teeth, the temporal arteries and the neck muscles.

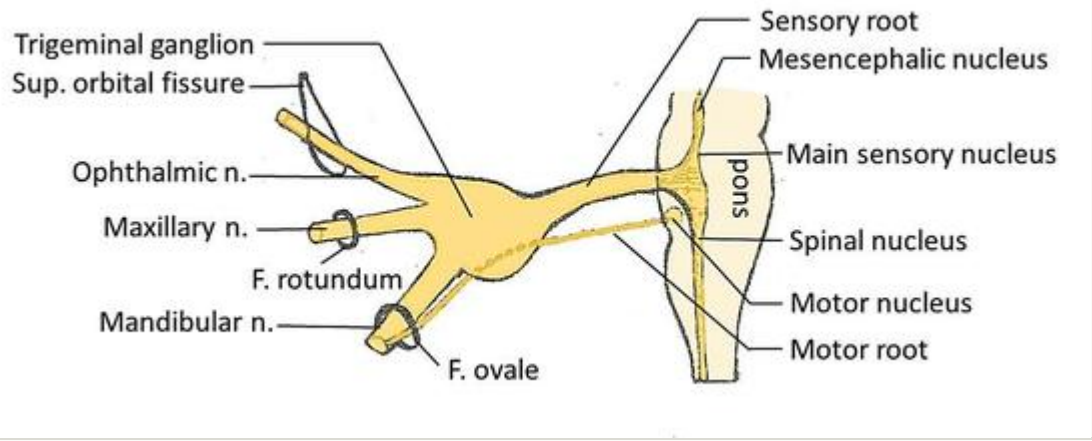


FIGURE 1 The trigeminal ganglion and its divisions, the pathway for most facial pain.
Anatomy atlas · Doss p. 148

2.2 Investigation

CT of the sinuses is not a screening test for facial pain, because incidental mucosal thickening is found in about a third of people without symptoms. It is used when malignancy, fungal disease or a complication is suspected, and in chronic rhinosinusitis that has failed medical treatment when surgery is planned. Dental radiographs and CT show odontogenic disease. MRI is used for trigeminal neuralgia, to look for a compressing vessel or demyelination, and for atypical or neurological features.

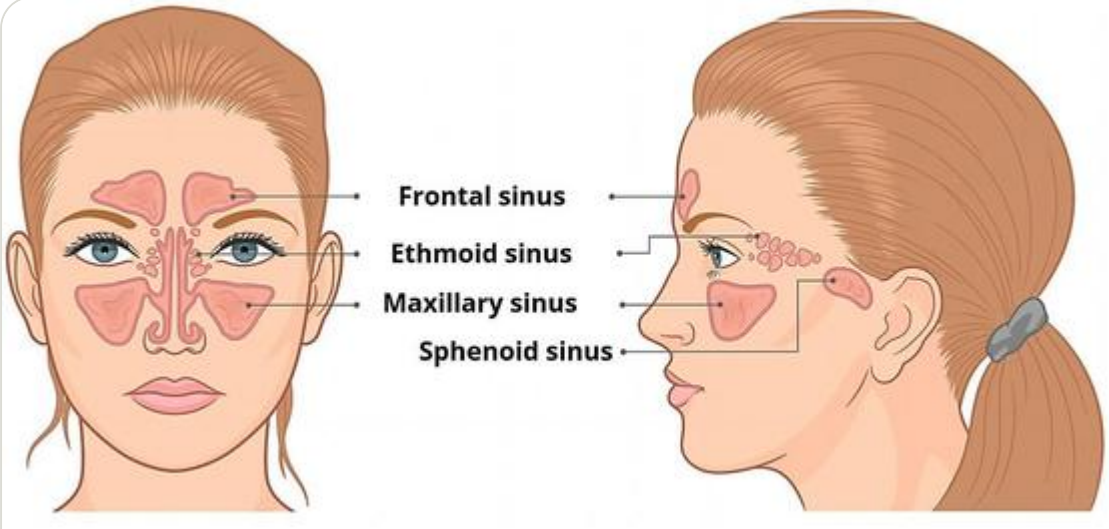


FIGURE 2 The paranasal sinuses.
Anatomy atlas · ENT Steps p. 126

3 CHAPTER THREE The causes

Recognising each.

FACIAL PAIN AND HEADACHE — FEATURES

CONDITION	PAIN AND TIMING	ASSOCIATED FEATURES	FIRST-LINE TREATMENT
Rhinosinusitis	Pressure with nasal symptoms; settles when they do	Blockage, discharge, smell loss; pus on endoscopy	Treat the sinuses
Migraine	Unilateral, throbbing, 4 to 72 hours	Nausea, photophobia, aura in about a quarter; may cause nasal congestion	Triptan acutely; propranolol, topiramate or amitriptyline to prevent
Tension-type headache	Bilateral, band-like pressing, mild to moderate	Few; stress, poor sleep	Simple analgesia, amitriptyline
Midfacial segment pain	Symmetrical pressure over the nasal bridge, cheeks or between the eyes	Normal endoscopy; hyperaesthesia of the skin	Low-dose amitriptyline
Cluster headache	Severe, orbital, 15 to 180 minutes, up to eight a day, in bouts	Ipsilateral tearing, rhinorrhoea, ptosis, restlessness; alcohol trigger	High-flow oxygen, subcutaneous sumatriptan; verapamil to prevent
Paroxysmal hemicrania	Like cluster but 2 to 30 minutes, many a day	Autonomic features	Indometacin (diagnostic response)
SUNCT and SUNA	Seconds to minutes, very frequent	Conjunctival injection and tearing	Lamotrigine
Trigeminal neuralgia	Electric shocks for seconds, usually V2 or V3	Triggered by touch, chewing, washing	Carbamazepine; MRI; microvascular decompression
Giant cell arteritis	Temporal, new, over 50	Jaw claudication, visual loss, raised ESR	Steroids at once

CONDITION	PAIN AND TIMING	ASSOCIATED FEATURES	FIRST-LINE TREATMENT
Temporomandibular disorder	Preauricular, with chewing	Clicking, tender joint and muscles	Soft diet, splint, physiotherapy

3.1 Midfacial segment pain

Midfacial segment pain is a form of tension-type headache affecting the middle third of the face. Patients describe a symmetrical pressure or tightness across the bridge of the nose, behind the eyes or over the cheeks, often feel their nose is blocked although it is not, and may have tender skin over the area. Endoscopy is normal, and the CT is normal or shows incidental changes. It responds to low-dose amitriptyline, starting at 10 mg at night and increased if needed, given for at least six weeks before its effect is judged and continued for about six months. Sinus surgery does not help.

3.2 Migraine and the nose

Migraine affects about one in seven people. Attacks can cause nasal congestion, discharge and tearing, and facial pain may be the only symptom, so migraine is commonly misdiagnosed as sinusitis. Acute attacks are treated with analgesics and triptans, used on no more than a few days a week to avoid medication-overuse headache. Prevention uses propranolol, topiramate (contraindicated in pregnancy) or amitriptyline, then candesartan, botulinum toxin for chronic migraine, or CGRP-targeted antibodies for refractory cases.

3.3 Contact point headache

The idea that a septal spur touching a turbinate causes pain is controversial. Most patients with contact points have no pain, and most with pain and contact points improve with medical treatment for midfacial segment pain. Surgery is considered only when a temporary local anaesthetic block abolishes the pain and other causes have been treated.

4 CHAPTER FOUR Pathway

Sinus or not?

FACIAL PAIN

1 Red flags?

- → **thunderclap, neurological signs, fever**
  emergency imaging
- → **over 50 with jaw claudication**
  ESR, CRP, steroids today
- → **numbness, diplopia, nasal mass**
  imaging and biopsy

2 Nasal symptoms with the pain?

- → **yes, pus or polyps on endoscopy**
  rhinosinusitis pathway, CT if surgery considered
- → **no nasal symptoms, normal endoscopy**
  step 3

3 Pattern of the pain

- → **unilateral throbbing with nausea**
  migraine
- → **symmetrical pressure over the midface**
  midfacial segment pain — amitriptyline
- → **short, severe, with ipsilateral autonomic features**
  cluster headache or another trigeminal autonomic cephalalgia
- → **electric shocks triggered by touch**
  trigeminal neuralgia — MRI, carbamazepine

4 Not improving

- → review analgesic overuse

→ refer to neurology or a pain clinic

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Why most "sinus headache" is not sinusitis:** pain needs nasal symptoms and endoscopic signs.
- **A structured pain history** with timing and associated features.
- **Midfacial segment pain:** features and amitriptyline.
- **The trigeminal autonomic cephalalgias:** cluster headache, paroxysmal hemicrania, SUNCT and SUNA.
- **Why CT is not a screening test** for facial pain.
- **Red flags,** especially giant cell arteritis and raised intracranial pressure.

5.2 Common pitfalls

COMMON PITFALLS

- Operating on the sinuses for facial pain with a normal endoscopy and incidental CT changes.
- Missing cluster headache because the patient also has a runny nose.
- Giving repeated antibiotics for recurrent "sinusitis" that is migraine.
- Missing medication-overuse headache.

Written for this app from Oxford Textbook of Otolaryngology ch. 31, Oxford Specialist Handbook ch. 3, Oxford Handbook of ENT ch. 3, Dhingra ch. 33. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

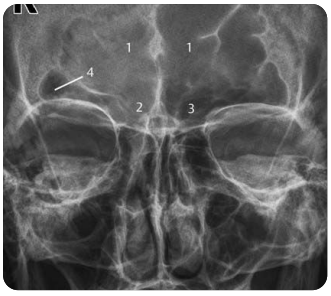
On imaging

The imaging of this topic: key findings, differentials and representative scans.

PARANASAL SINUSES · INFECTION & INFLAMMATION

Frontal Sinusitis

- The anatomy and pathology of the frontal recess required special attention.
- In cases of persisting drainage problems in the region of the frontal recess, the septum between the frontal sinuses may be removed for establishing drainage by way of the left frontal recess.
- Bony destruction with infiltrative characteristics is suggestive of malignancy.
- Calcifications in the opacified area are suggestive of fungal infections.



Plain radiograph, Caldwell. It is difficult to evaluate the superior aspect of the frontal sinuses (1) to determine whether these are opacified or are…

ORBIT & SINUS RELATIONS · INFECTION & INFLAMMATION

Mucocele

IMAGING FINDINGS

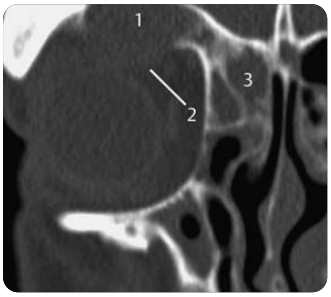
- Expanded, airless sinus with smooth remodelled walls
- Variable density and signal with protein content
- Rim enhancement only

TOP DIFFERENTIALS

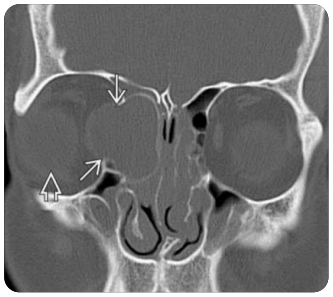
- Allergic fungal sinusitis
- Polyposis
- Tumour (solid enhancement, destruction)

CHECKLIST

- Orbit and skull base erosion
- Frontal recess obstruction as the cause



CT, coronal. The left frontal sinus is opacified with expansion of its inferior bony outline through the roof of the orbit (1). Although this is a bone-…



Coronal bone CT shows the typical features of a right ethmoid mucocele. The sinus is opacified, and there is remodeling of the surrounding bony…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 716.

PARANASAL SINUSES · TUMOURS

Sinonasal tumours

IMAGING FINDINGS

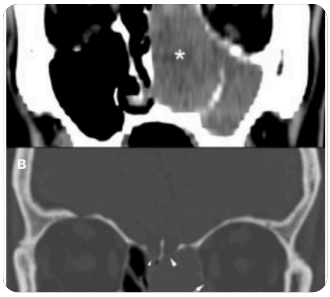
- Unilateral solid mass with bone destruction
- Intermediate T2, solid enhancement, distinct from bright secretions
- Spread to orbit, skull base, pterygopalatine fossa, cheek

TOP DIFFERENTIALS

- Inverted papilloma
- Lymphoma
- Olfactory neuroblastoma, adenocarcinoma, melanoma

CHECKLIST

- Periorbita and dura
- Perineural spread along V2 and the vidian nerve; nodes



Coronal reformatted images from a noncontrast CT in soft tissue window (a) and bone window settings (b). The mass (asterisk) causes resorption of…



Axial CECT shows the typical location and appearance of an antral SCCa. There is extension into the premaxillary soft tissues anteriorly an…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 754.